A study of the effect of grid quality in simulated flow around square cylinder

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1. Objectives

The flow around a square cylinder has been widely used in literature, both as a simple setup producing complex phenomena which can be experimentally observed, both as a method to assess the quality of simulated flows and the effect of several parameters of the setup, as testified by benchmarks in *ERCOFTAC* [1]. However, the effect of grid quality on simulated flows has often been overlooked, and we aim at progressing towards that way.
We take cell skewness as a metric for grid quality, and we confront simulations made with two widely used yet very different grid boundary layers, differing mainly for the distribution of skewness, with more accurate simulations and experimental results in literature.
By comparing different results, such as integral parameters and distribution of velocities, we aim at assessing the effect of these different grids on simulation and the discrepancies with both more accurate simulations and experimental data, and at finding physical interpretation and numerical justification of these discrepancies between different simulations and especially with experimental data.
The case of the flow around a square cylinder has been chosen also because the Strouhal frequency is almost independent from the Reynolds number, for approximately 1.0e+3 $< Re <$ 1.0e+7. In any case, the test case on *ERCOFTAC* is carried with $Re =$ 2.2e+4, following the pioneering work of Lyn [2].

2. Computational model

2.1. Turbulence model

The classical Unsteady Reynolds Averaged Navier-Stokes (URANS) equations are used to model the incompressible, turbulent, separated, unsteady flow around the square section:

$$\begin{cases} U_{i,i} = 0 \\ U_{i,t} + U_j U_{i,j} = -\frac{1}{\rho} P_{,i} + \left(\nu \left(U_{i,j} + U_{j,i}\right) - R_{ij}\right)_{,j} \end{cases}$$

where U_i and P are the averaged $i-$velocity and pressure, ρ the air density, ν its kinematic viscosity and the Reynolds stresses R_{ij} are expressed by the Boussinesq assumption as $R_{ij} = -2\nu_t S_{ij}$, where S_{ij} is the mean field strain rate, and $\nu_t = C_\mu \frac{k^2}{\varepsilon}$ is the turbulent viscosity expressed in terms of the turbulent kinetic energy k and its dissipation rate ε. The $k-\varepsilon$ RNG turbulence model is used to close the URANS equations. For the near-wall treatment a non-equilibrium wall function approach is used. The following boundary conditions are imposed:
• Dirichlet for both velocity field and turbulent variables at the inlet
• same Dirichlet for turbulent variables and no-stress condition at the outlet
• periodic on lower and upper boundaries
• no-slip condition at cylinder wall

2.2. Numerical approach

CDS is used for the diffusion term, QUICK for the convective terms in the U, k and ε equations. Second-order implicit method is used for time discretization, and the PISO algorithm for pressure-velocity coupling.

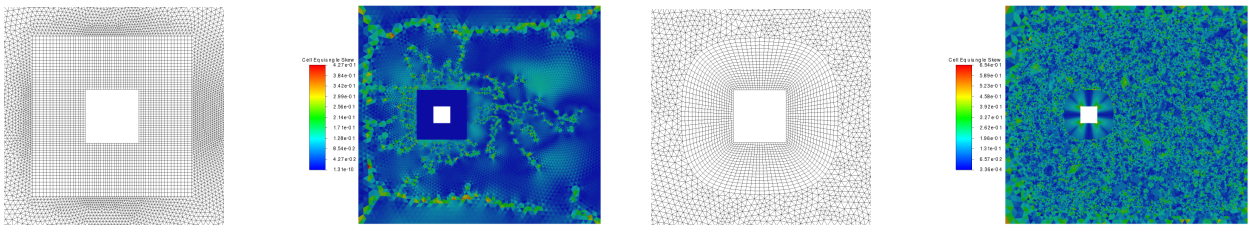

Figure 1: Grid boundary layer and skewness distribution for the two grids

3. Setup of the study

In according to [1] the incoming flow is characterized by $Re = 2.2e{+}4$, incidence $\alpha = 0$, turbulence intensity $It = 2\%$ and turbulence length scale $Lt = 0.5D$. Dimensionless time-step is set to $\Delta t = 0.02$ to account for the CFL condition and simulation are extended over a whole period starting from a fully developed flow.
Both grids are overall hybrid, consisting of an outer zone discretized by structured quadrangular cells, and an inner zone discretized by unstructured triangular cells. The main difference between the two overall grids is the grid boundary layer in proximity of the cylinder: in the first case we consider a structured, fully orthogonal, equispaced grid (*Full Ortho*); in the second case we consider a structured O-Type, boundary-fitted, radial, non-equispaced grid (*Radial*) (see Fig. 1).
The main differences between GBLs are their skewness distribution and the effect on grid density just outside the GBL. While the *Full Ortho* GBL adds virtually no grid error, since it has a null distribution of skewness, the *Radial* GBL presents skewed cells loosely distributed, especially over the edge of the cylinder.
In addition to this, in order to have a fair comparison, both grids discretize the edges of the cylinder with 24 cells, such that requirements for wall-function approach are met. Thanks to this, the *Full Ortho* GBL, since it has no grid gradient, forces a high grid density close to the boundary between GBL and triangular mesh, and thus surely improving accuracy in the near wake region. On the contrary, the *Radial* GBL causes an overall zero grid gradient in the triangular mesh.

4. Results

The present results are compared both among them and with some experimental ([3, 4, 5, 6, 7, 2]) and numerical ([8, 9]) results from literature which are supposed to be of high-quality and therefore considered as reference. Our results are especially compared with [2], since it is one of the few experimental tests with phase-resolved results, which we have also carried. In particular, following the same approach and dividing a period in 20 phases, we have compared results at first phase (where C_L is supposed to be barely positive and increasing) and at ninth phase (where C_L is supposed to be barely positive and decreasing).

4.1. Flow fields

We begin by comparing the flow lines in the region near the cylinder, in order to preliminarily assess the quality of the simulation (see Fig.2 and Fig.3) . In both phases, both grids seem to capture well the vortex shedding phenomenon. However, in phase 1, the *Full Ortho* GBL captures the re-circulation zone in the upper part of the cylinder better than the *Radial* GBL, which captures only the flow separation, due to the Reynolds number being reduced by artificial viscosity, in particular skewness viscosity. This also happens in phase 9 with the re-circulation zone in the lower part of the cylinder.
Moving on to vorticity magnitude isocontours, both simulation grasp the nature of Von Kármán like vortices along the wake, but with some small differences: the longitudinal spacing between successive vortices decreases by approximately 0.1-0.3D, as found in [10]; moreover, the *Radial* GBL also underestimates vorticity on the upwind angles of the cylinder, due to the unnecessary cell skewness in the upwind part of the GBL. In fact, as stated in [10], it is possible to improve accuracy by placing a structured fully orthogonal grid in the upwind part of the GBL, and joining it properly with the downwind GBL.

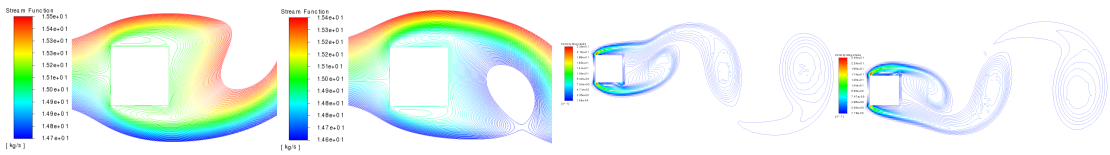

Figure 2: Flow lines and vorticity isocontours in phase 1 and 9 for Full Ortho grid

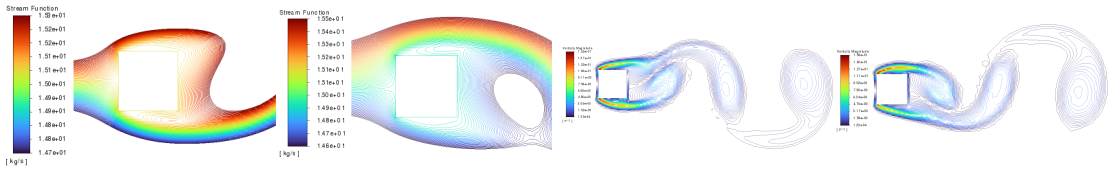

Figure 3: Flow lines and vorticity isocontours in phase 1 and 9 for Radial grid

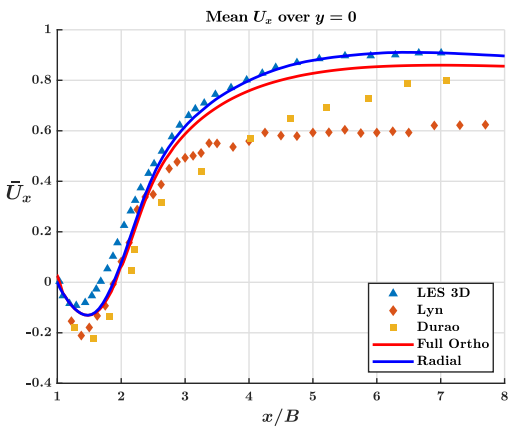

Figure 4: Mean X-wise velocity along $y = 0$

We compare now the distribution of velocities in both grids with results in literature, starting with the stream-wise velocity U_x (which is dimensionless in our case since we consider $U_\infty = 1$) along the wake $y = 0$, averaged in time over a period, shown in Fig.4.

In the re-circulation zone, the experimental values are almost twice the simulated ones, and surprisingly the 3D LES performs worse than our model, by underestimating both re-circulation length and the velocity defect. The biggest difference occurs further in the wake: the experimental values, although showing different behaviours, predict lower far-wake velocity than the simulated ones, probably due to both the 2D model and the poor modelization of turbulence. In fact, the results in [9], performed with a 3D DNS, fit experimental values better.

Looking at the phase-resolved results, we can compare the velocities in the two chosen phases. The y-velocity in wake (along $y = 0$), as expected, oscillates coherently with the vortex-shedding (see Fig.5). Our simulations however, model poorly the velocity dissipation further down the wake, which is evident in experimental data. This is probably due to the turbulence model adopted, known to underestimate dissipation, and might be improved by more realistic models such as $k - \omega$ or LES. Some non-physical peaks are also visible in the simulations, which might be caused by some unidentified source of dispersion error. The results with both GBLs are quite similar except for a seemingly small, yet constant phase delay: surprisingly, this makes the *Radial* GBL perform better, while also making the velocity along each vortex slightly higher than the one simulated with the *Full Ortho* GBL.

Moreover, in phase 1 both simulations dramatically overestimate both the velocity magnitude and the instantaneous re-circulation length near the near-wake region, which is a peculiar behaviour since this overestimation does not appear in phase 9.

We can also look at the x-wise velocity along $x = 1$, that is at the downwind edge of the cylinder, in Fig.6. A noteworthy fact concerns the two re-circulation zones near to the cylinder's corners. Experimental data predicts both in phase 1 and 9 the presence of these zones, one of which is slightly lower than the other. Our simulations seem to be able to detect only the bigger zone while they completely miss the smaller one. Moreover, the simulation with the *Radial* GBL, after a small initial zone of numeric instability near to the corners of the cylinder, captures well the experimental behaviour far from the cylinder, like the *Full Ortho* GBL.

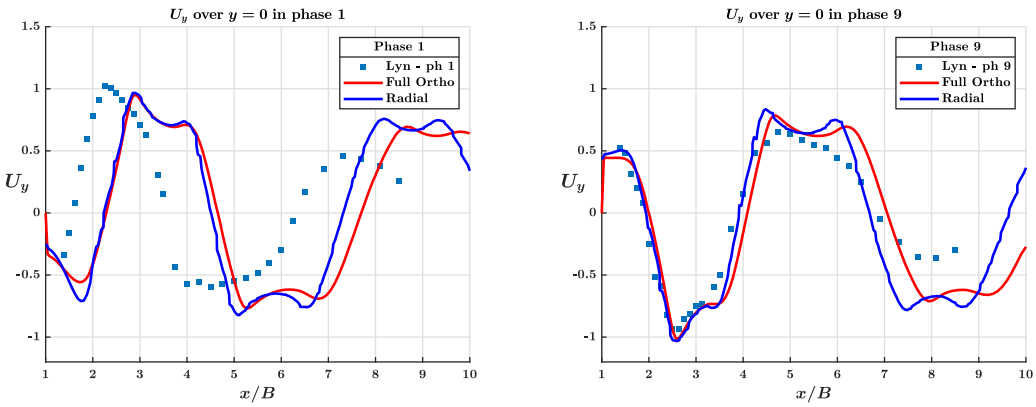

Figure 5: Y-wise velocity along $y = 0$ in phase 1 and phase 9

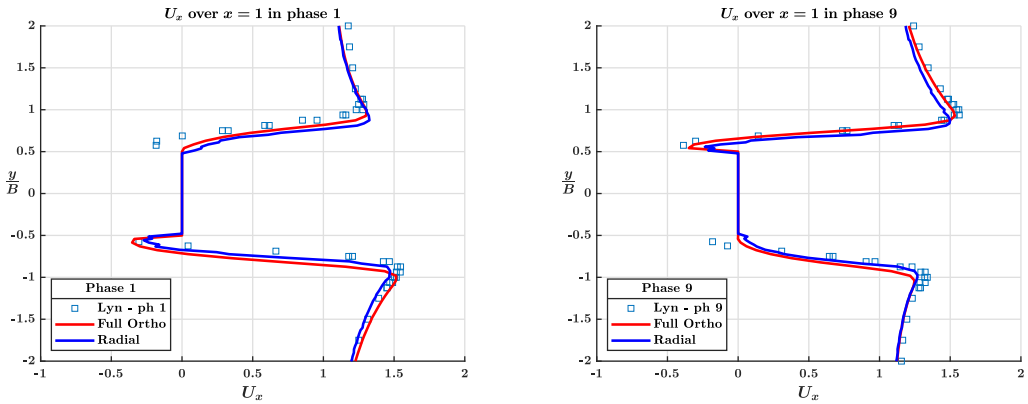

Figure 6: X-wise velocity along $x = 1$ in phase 1 and phase 9

4.2. Stress distribution at wall

We can now look at pressure distribution over the upper half of the cylinder. In this analysis it is important to notice that results are highly scattered and neither high-fidelity simulations nor experimental data can lead to quantitatively accurate conclusions, especially concerning the standard deviation of C_p. Therefore we should focus our attention mainly on qualitative aspects.
From Fig.7 we can assert that present simulations have good results on the upwind edge (i.e. in the $(0, 0.5)$ interval) while they drastically differ from reference results after 0.5, especially for the standard deviation of C_p.
Both graphs evidently show that the *Radial* GBL has worse results than the *Full Ortho* GBL. In fact, concerning the estimate of the mean value of C_p on the interval $(0.5, 1.5)$, while the latter is essentially coherent (also quantitatively) with other numerical simulations and experimental results, the latter underestimates the minimum near 0.5 and then overestimates the value of the plateau.
It is remarkable that the *Radial* GBL represents a bound (respectively upper for the mean value and lower for the standard deviation in Fig.7) for the ensemble of data. This means that the simulation with *Radial* GBL overestimates the mean value of C_p and underestimate its standard deviation.

4.3. Bulk forces

Finally, looking at integral parameters we can assess the total forces acting on the cylinder.

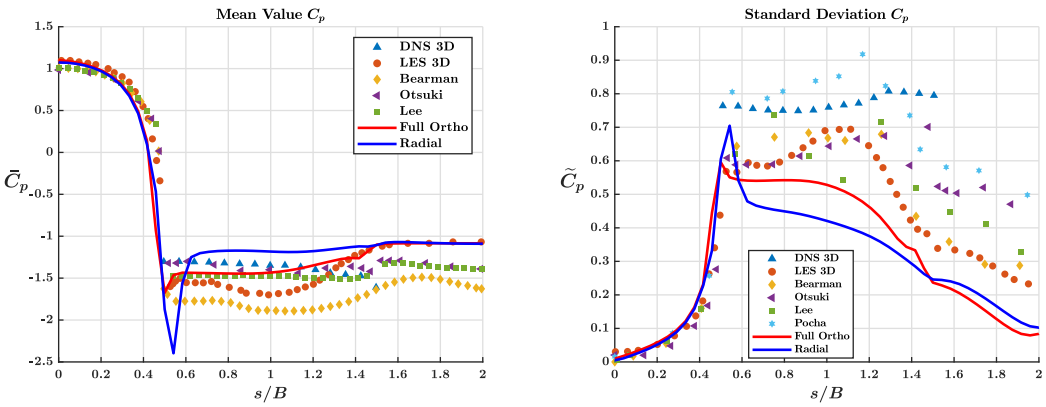

Figure 7: Mean value and standard deviation of Pressure Coefficient across upper half of cylinder

Test	St	$\overline{C_D}$	$\widetilde{C_D}$	$\widetilde{C_L}$
Lyn [2]	0.135	2.05 - 2.23	n.a.	n.a.
Sakamoto [7]	0.134	2.22	0.132	1.45
Durao [11]	0.139	n.a.	n.a	n.a
Murakami [8]	0.132	2.09	0.13	1.60
Full Ortho	0.138	1.90	0.056	1.39
Radial	0.151	1.73	0.083	1.18

There is a great variance in results, both numerical and experimental; it is clear however that, while the *Full Ortho* GBL predicts an admissible Strouhal number, the *Radial* GBL predicts a higher frequency. In strict terms, both simulations (see Fig.8) predict a C_L frequency equal to Strouhal number, and a C_D frequency equal to two times the Strouhal number, in accordance with past experimental results. The Strouhal number for *Radial* GBL, although higher than the mean experimental value, falls in between the St with *Full Ortho* GBL and the St simulated in [10] with *O-type* GBL. This repeats as well with the other bulk parameters depicted in the table. This is consistent, since thanks to numerical errors and skewness-induced errors, St increases while mean C_D and fluctuations of C_L decrease, due to the increase in effective viscosity. Against the trend however, the C_D fluctuations for the *Radial* GBL are closer to experimental values than the *Full Ortho* GBL ones. In any case, it is necessary to observe that just a simple alteration of the grid might cause a dramatic underestimation of the forces acting on the structure.

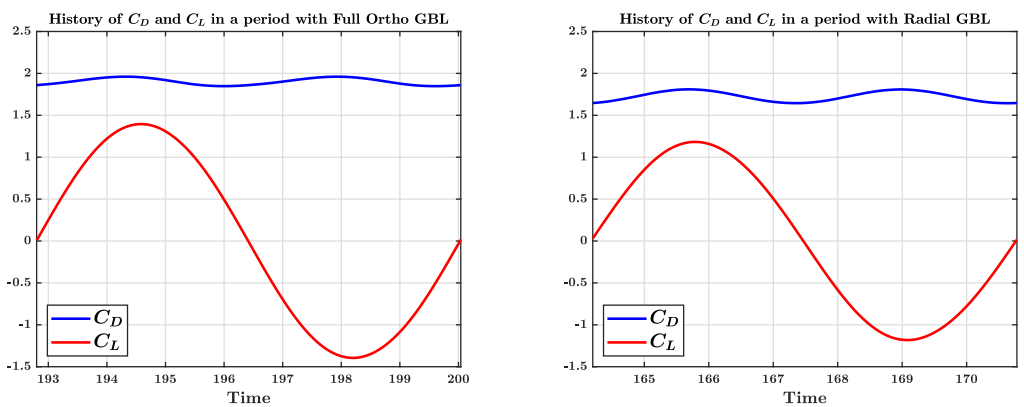

Figure 8: History of Drag and Lift coefficient across a period

5. Conclusions

Referring to the *Radial* GBL we can conclude that the presence of moderately skewed cells (or better to say, skewed faces) in a relatively small but crucial zone of the domain (i.e. a neighbourhood of the upwind corners of the cylinder) produces errors which are then transferred by convection in the wake that, in the end, may modify subtly, yet considerably the overall flow field. A possible way to understand how this grid error affects the solution is considering that it is a viscous-type error. In fact the artificial viscosity added due to skewness around the separation point increases the actual viscosity $\nu + \nu_t + \nu_{sk}$ and therefore reduces the simulated Re number. This conclusion can be drawn from the results presented in 4.3. The simulation with the *Radial* GBL led to an estimate of Strouhal number, C_D average and C_L fluctuations which are related to a flow with Re lower than the one intended to simulate.
The former analysis is also supported by the comparison between the two used grids. Despite the same numerical setup and the a-priori same quality, the *Full Ortho* GBL led to results that are closer to the high-quality ones. Since the only main difference between these two GBLs is the skewness distribution, we can infer that this is the cause for the worse results obtained with the *Radial* GBL.
In the end we want to remark that almost all of our result show that 2D simulations hardly lead to satisfying estimates. Therefore we think further investigations on the relative magnitude between skewness-induced errors and dimension reduction errors, as well as for other turbulence model choices, are needed.

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